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70H-1024600

Overview

The 70H-1024600 series display, designed by Waveshare Team, is for customers who need a lightweight HDMI display. As it is thinner and lighter than any traditional HDMI display, users can do the second development by themselves easily.
For simplifying the display, we integrate the power supply, display, touch, and control signal as the 40PIN multi-functional header. Just need an FPC cable, you can lower the project cost with the thinnest display, and apply it in any embedded projects.

Feature

- 7-inch IPS/QLED screen, 1024 x 600 hardware resolution.
- Capacitive touch, support five-point touch (this function is only for the touch version).
- COF thin and light structure, providing 40PIN interface one-line communication, integrated development is more convenient.
- High compatibility, when used with development accessories, it supports main control boards such as Raspberry Pi, PC, and Jetson nano.
- Support backlight control, more power saving.

Size	Resolution	Display Panel	Viewing Angle	Touch Type
7"	1024×600	IPS/QLED	170°	Capacitive(optional)

Touch Points	Consumption	Certification 1	Certification 2
5-Points(optional)	Low Power	RoHS2.0	CE

Specification

Item	Description	Unit
Product Model	70H-1024600 Series	
Size	7.0	Inch
Viewing Angle	170	Deg
Resolution	1024 (RGB) × 600	Pixels
Touch Screen Size	173 (H) × 103 (V) × 4.2 (D)	mm
Display Size	164 (H) × 97 (V) × 3.5 (D)	mm
Display Area	154.2144 (W) × 85.92 (H)	mm
Pixel Pitch	0.1506 (H) x 0.1432 (V)	mm
IPS Version Color Gamut	45%NTSC	
QLED Version Color Gamut	78%NTSC	
Brightness	350	cd/m²
Backlight Adjustment Method	PWM software backlight adjustment	
Contrast	800:1	
Color Depth	16.7M	
Refresh Rate	60	Hz
User Interface	40PIN 0.5mm FPC	
Weight	Note1	g
Power Consumption	Note2	Watt

- **Note 1:** 70H-1024600-IPS=94g, 70H-1024600-IPS-CT=153g, 70H-1024600-QLED=106g, 70H-1024600-QLED-CT=165g.
- **Note 2:** 70H-1024600-IPS=2.15W, 70H-1024600-IPS-CT=2.3W, 70H-1024600-QLED=2.15W, 70H-1024600-QLED-CT = 2.3W.

Electrical Specifications

Parameter	Minimum value	Standard value	Max value	Unit	Note
Input Voltage	4.75	5.00	5.25	V	Note 1
Input Current	500	500	TBD	mA	Note 2
Output Voltage	3.1	3.3	3.5	V	Note 1
Output Current	0	200	200	mA	Note 3
Operating temperature	-10	25	70	℃	Note 4
Storage temperature	-20	25	80	℃	Note 4
Operating Humidity	10	60	90	%RH	Note 4

- **Note 1:** Input voltage/output voltage exceeding the maximum value or improper operation may cause permanent damage to the device.
- **Note 2:** Input current/output current must be greater than 500mA, otherwise it will cause startup failure or abnormal display, and it may cause permanent damage to the device if it is in an abnormal state for a long time.
- **Note 3:** The output current needs to be less than 200mA, otherwise it will fail to be started or display abnormally, and it may cause permanent damage to the device if it is in an abnormal state for a long time.
- **Note 4:** Please do not store the display in a high-temperature and high-humidity environment for a long time. The display must work within the limited value range, otherwise, the display may be damaged.

Interface Definition

PIN	Logo	Function Description
1~3	NC	Not connect
4~6	5V IN	DC 5V power input
7	3V3 OUT	DC 3.3V power output
8~10	GND	Ground (0V)
11	TMDS Data2+	Differential data signal
12	GND	Ground (0V)
13	TMDS Data2-	Differential data signal
14	TMDS Data1+	Differential data signal
15	GND	Ground (0V)
16	TMDS Data1-	Differential data signal
17	TMDS Data0+	Differential data signal
18	GND	Ground (0V)
19	TMDS Data0-	Differential data signal
20	TMDS Clock+	Differential data signal
21	GND	Ground (0V)
22	TMDS Clock-	Differential data signal
23	CEC	Consumer electronics control signals
24	SCL	I2C clock line, internal 10kΩ pull-up
25	SDA	I2C data line, internal 10kΩ pull-up
26	GND	Ground (0V)
27	+5V Power	With HPD to achieve insertion detection
28	Hot Plug Detect	Hot plug detect signal
29~32	NC	Not Connect
33	D+	USB differential data signal
34	D-	USB differential data signal
35	GND	Ground (0V)
36	BL_PWM	Display backlight adjustment
37	BL_EN	Display backlight enable
38~40	NC	Not connect

Adapter Board Design

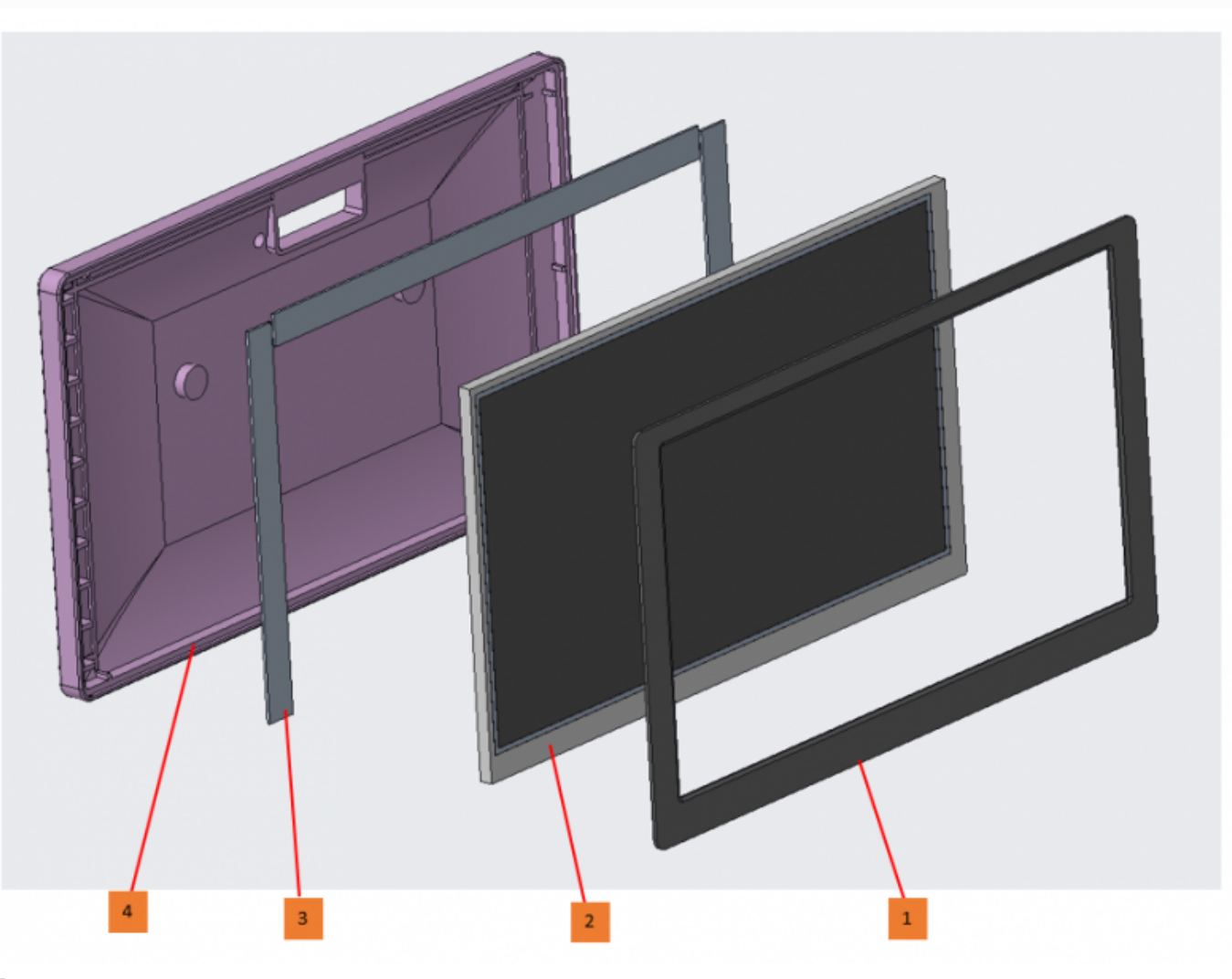
You can refer to adapter board schematic to integrate the display, please note:

- HDMI needs to use an equal-length differential cable, and the differential impedance of the four pairs of differential signals needs to be controlled at 100Ω±10%.
- The four-pair differential signal cables of HDMI are required to be less than 5 inches.
- The length deviation within the HDMI differential pair is required to be controlled within 20mil, and the maximum shall not exceed 25mil.
- The interface ESD device needs to be placed close to the HDMI socket.
- The 5V power input line needs a wide enough path, it is recommended to exceed 30mil to ensure that the voltage input will not have an excessive loss.
- Pay attention to the line and direction to prevent reverse connection.

Structure Design

NOTE: The assembly materials used below are not included in the shipped kit! ! This is only for reference.

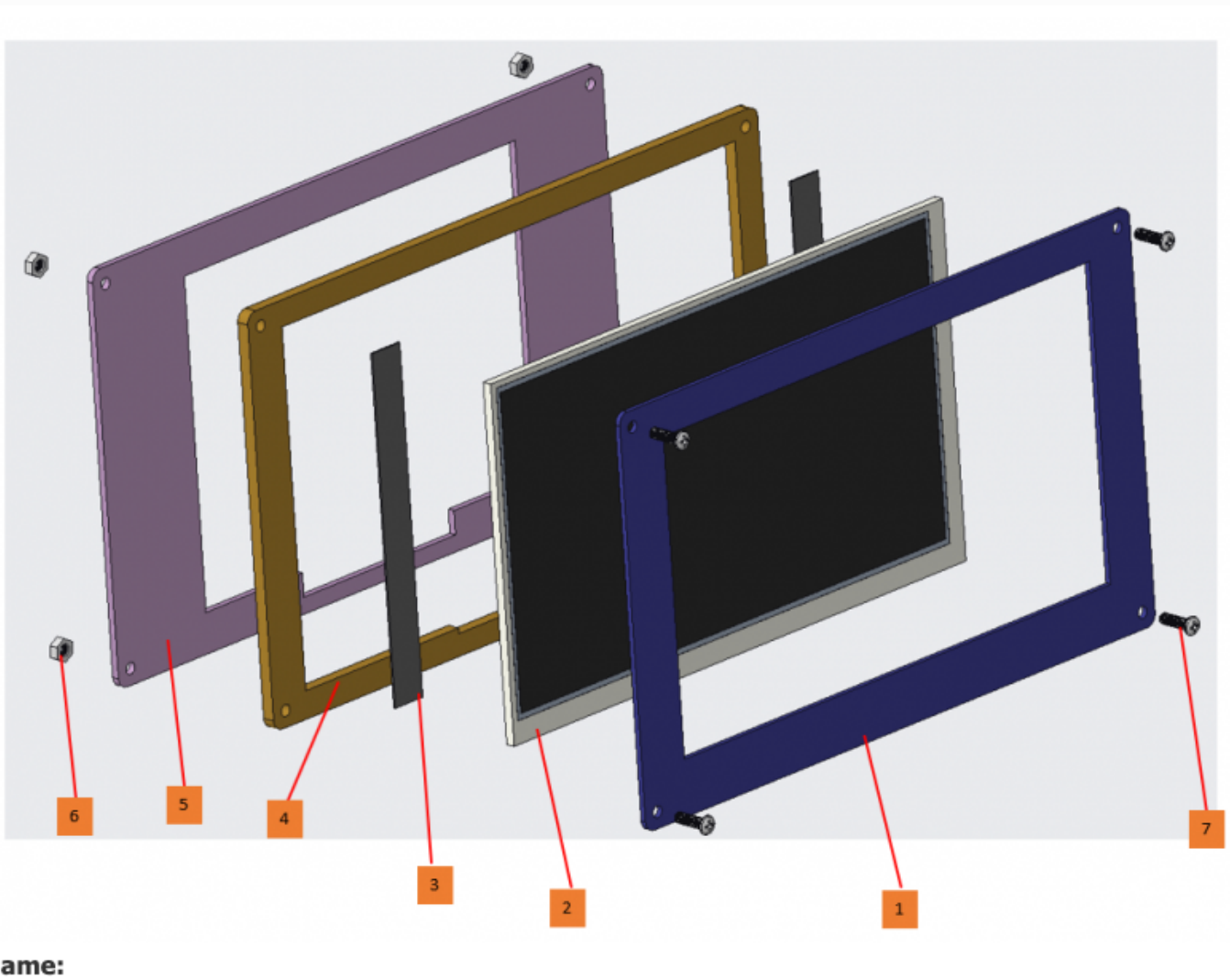
3D Print Shell Solution



Name:
1. The front panel comes with adhesive backing
2. 70H-1024600-IPS/QLED
3. Adhesive foam
4. Rear case

Assembly Instructions:
Affix the back of the display screen (part 2) with a suitable thickness of adhesive foam (part 3), then directly embed the display screen (part 2) into the rear case (part 4), and then paste the front panel (part 1) to the corresponding location.

How to Fix the Acrylic

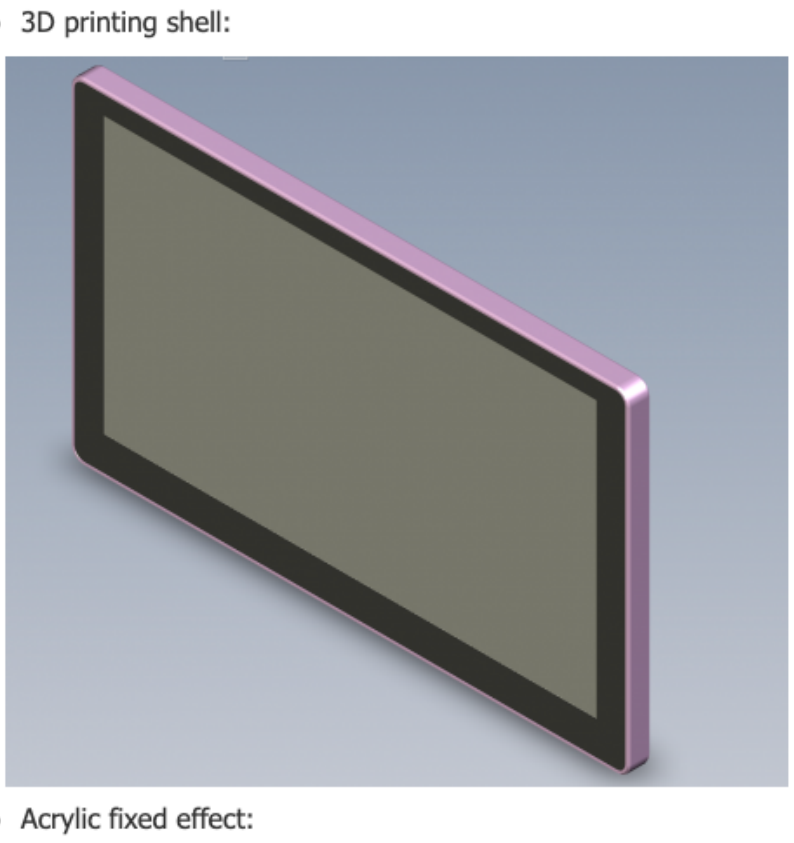


Name:
1. Front panel (1.5mm thick acrylic board is recommended)
2. 70H-1024600-IPS/QLED
3. Adhesive foam
4. Medium plate (It is recommended to use an acrylic plate with a thickness of more than 3.5mm)
5. Bottom plate (It is recommended to use an acrylic plate with a thickness of more than 2mm)
6. Nuts
7. Pan head screws

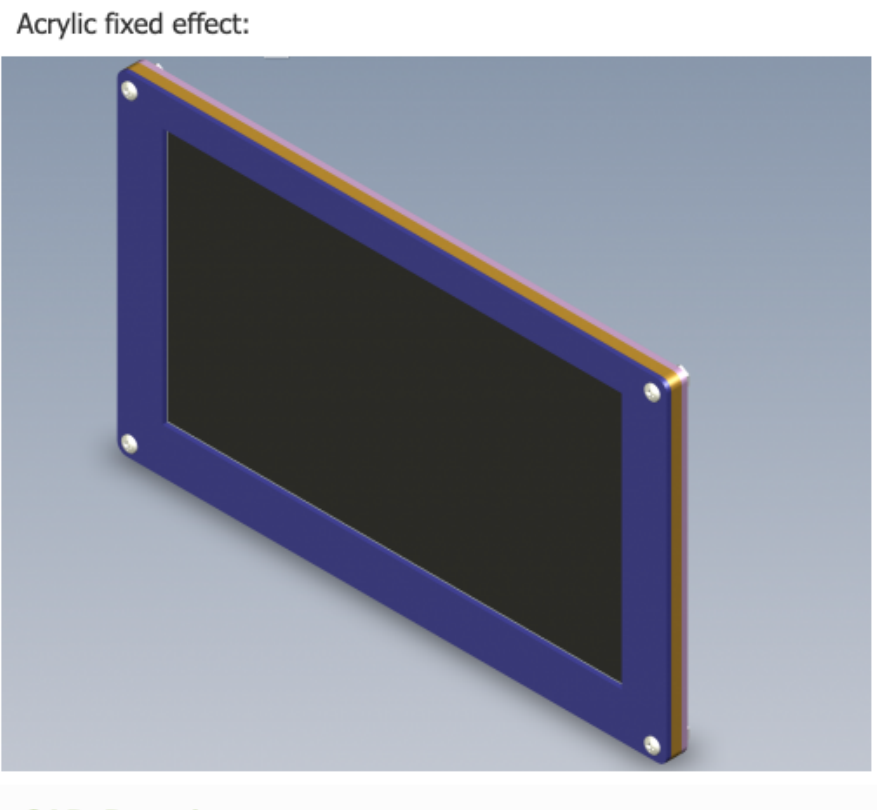
Assembly Instructions:
Align the front panel (part 1) and the midplane (part 4) and thread the screws (part 7) into the display (part 2) Paste a suitable thickness of adhesive foam (part 3) on the back of the display screen, then install the bottom plate (part 5), and finally lock the nut (part 6).
Note: The LCD display surface cannot be installed parallel to the ground, otherwise it will fall off under the action of gravity. In addition, when it needs to be installed at an angle, the back of the LCD screen should keep at least an angle of less than 75 degrees with the horizontal direction.

Assembly 3D Drawing

- 3D printing shell:



- Acrylic fixed effect:



CAD Drawing

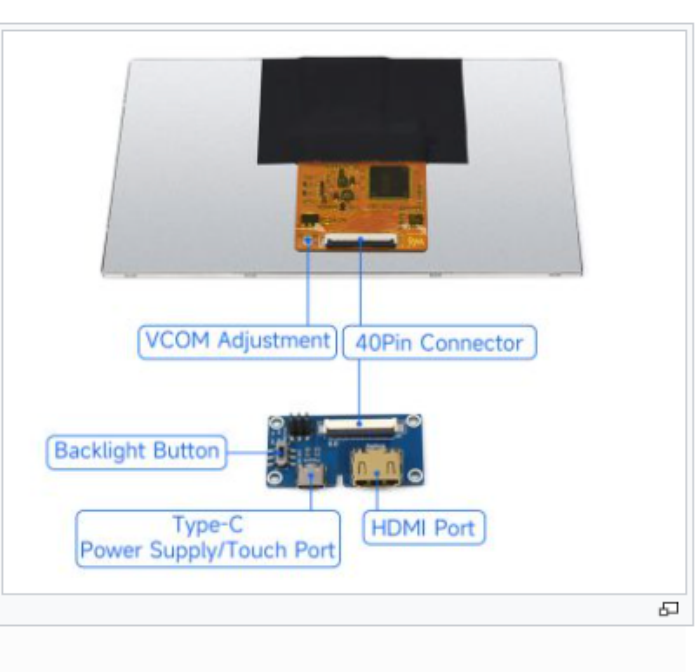
- 70H-1024600-IPS-CAD
- 70H-1024600-QLED-CAD

EDID Timing Parameters

If the system of the main control board can automatically identify the EDID for display, there is no need to set additional timing parameters. Otherwise, you can refer to the following EDID settings:

Pixel Clock	H Adressable	H Blanking	V Adressable	V Blanking	H Front Porch	H Sync Width	V Front Porch	V Sync Width	H Image Size	V Image Size	H Border	V Border
49.00	1024	288	600	24	48	96	3	10	154	86	0	0

How to Develop

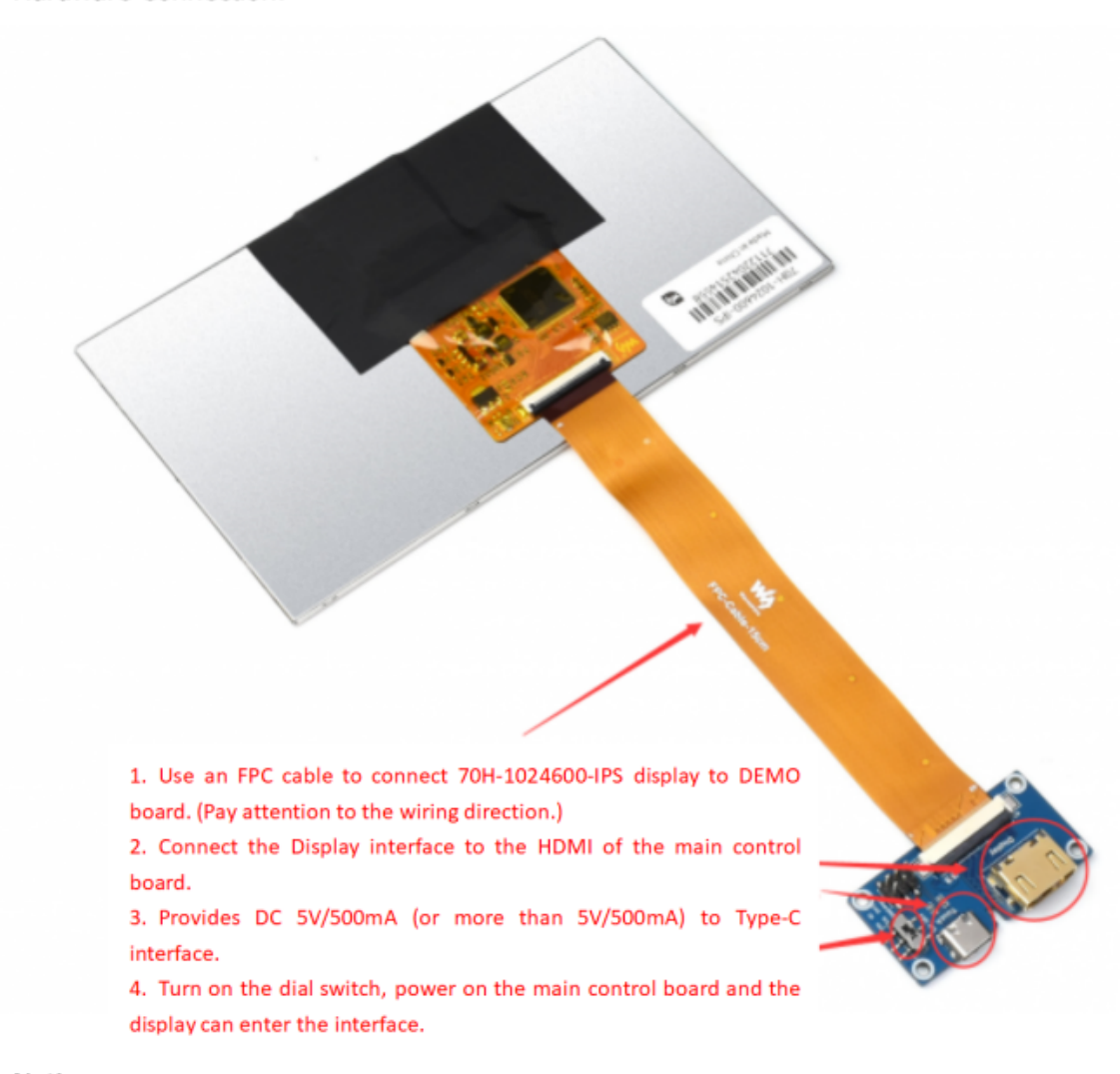


- If you are using the Waveshare 70H-1024600 series touch display screen for the first time, it is recommended to use it with the development accessories:

Adapter board pin header definition	Function introduction
5V	DC 5V power input
3V3	DC 3.3V power output
GND	Ground (0V)
EN	Backlight enable
PWM	Backlight adjustment
IO	NC

Working with PC

- Support Windows 11/10/8.1/8/7 system. Instructions:
  1. Use the FPC-Cable-15cm to connect the 70H-1024600 display screen to the development accessories (pay attention to the cable direction, 1 to 1, 40 to 40 serial number connection).
  2. Connect the Display interface to the PC HDMI interface.
  3. Connect the Touch interface to the USB interface of the PC.
- Hardware Connection:

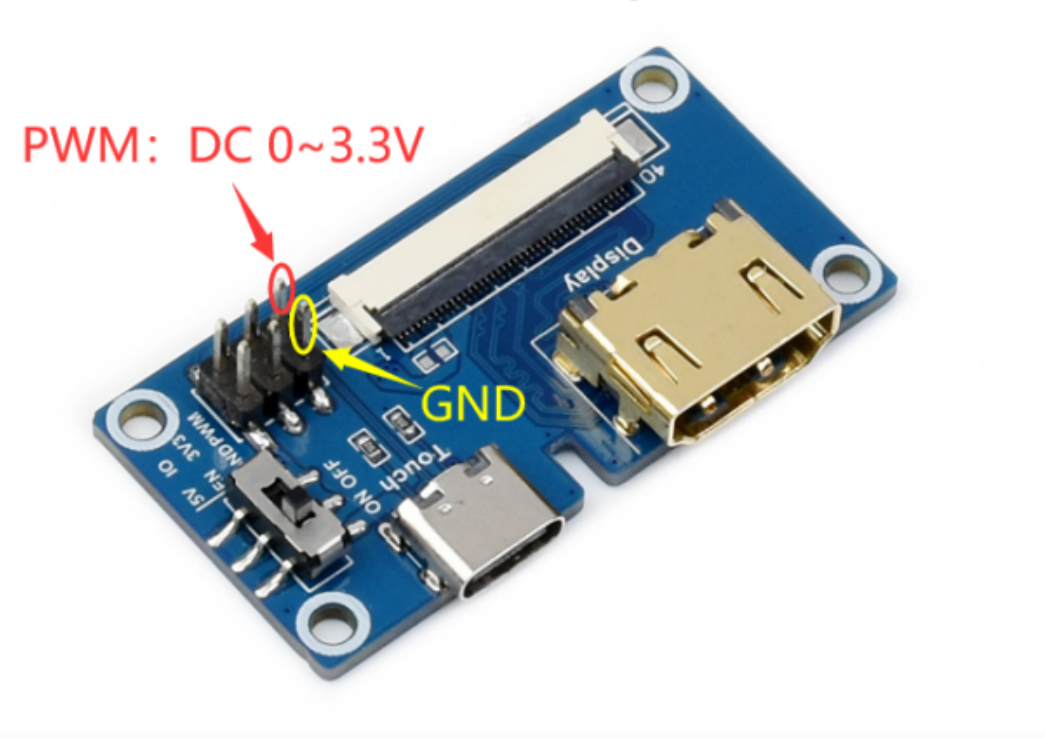


1. Use an FPC cable to connect 70H-1024600-IPS display to DEMO board. (Pay attention to the wiring direction.)
2. Connect the Display interface to the HDMI of the main control board.
3. Provides DC 5V/500mA (or more than 5V/500mA) to Type-C interface.
4. Turn on the dial switch, power on the main control board and the display can enter the interface.

- Notice:
- Some PCs do not support plug-and-play with HDMI devices. Generally, they can be used normally after restarting the system.
- Insufficient power supply of the USB interface of some PCs will cause the LCD screen to flicker, which can be restored to normal use after connecting to an external 5V/1A power supply.

PWM Light Adjustment

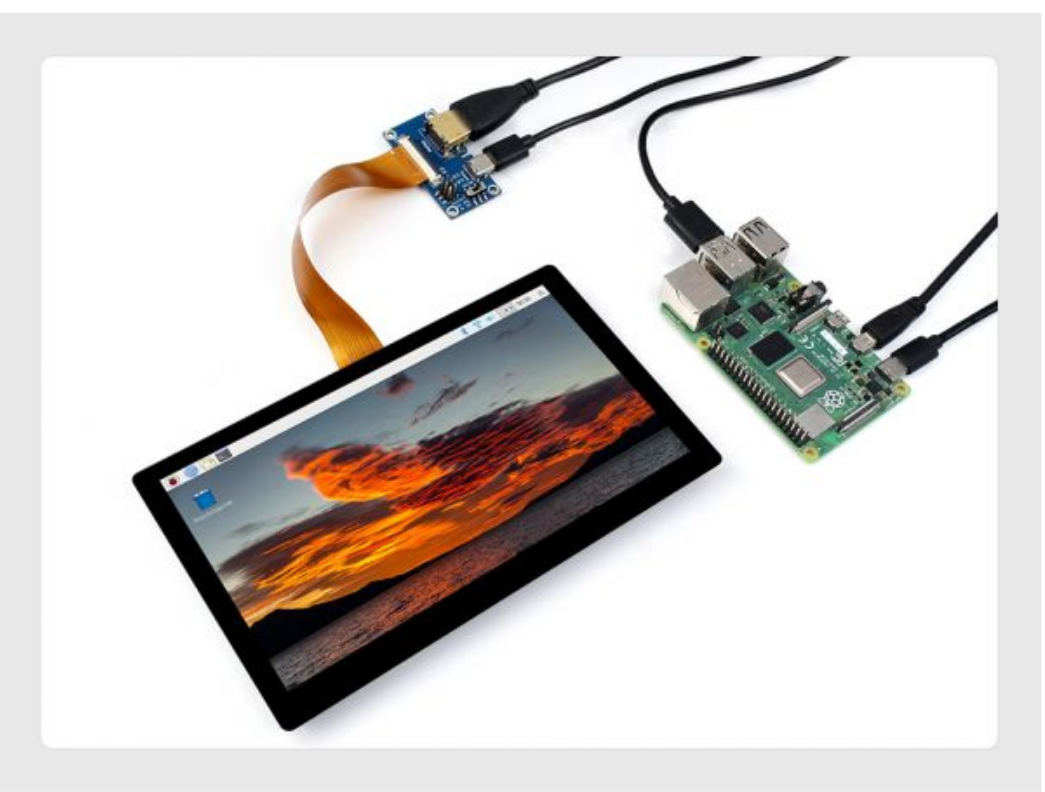
Connect the GND of the microcontroller (such as a potentiometer) to the GND pin in the figure below, and connect the voltage output terminal (DC 0~3.3V) of the microcontroller to the PWM pin as shown in the figure below so as to control the backlight brightness by changing the voltage.
Note: DC 3.3V is the darkest and 0V is the brightest.



Working with Raspberry Pi

Hardware Connection

1. Use the FPC-Cable-15cm to connect the 70H-1024600 touch screen to the Demo board (pay attention to the cable direction, 1 to 1, 40 to 40 serial number connection).
2. Connect the Display port to the Raspberry Pi HDMI port.
3. Connect the Touch port to the USB port of the Raspberry Pi.



Software Setting

- Support Raspberry Pi OS / Ubuntu / Kali and Retropie systems.When the LCD works on systems such as Raspberry Pi, the resolution must be set manually, otherwise, the display resolution will be incorrect and the experience will be affected.

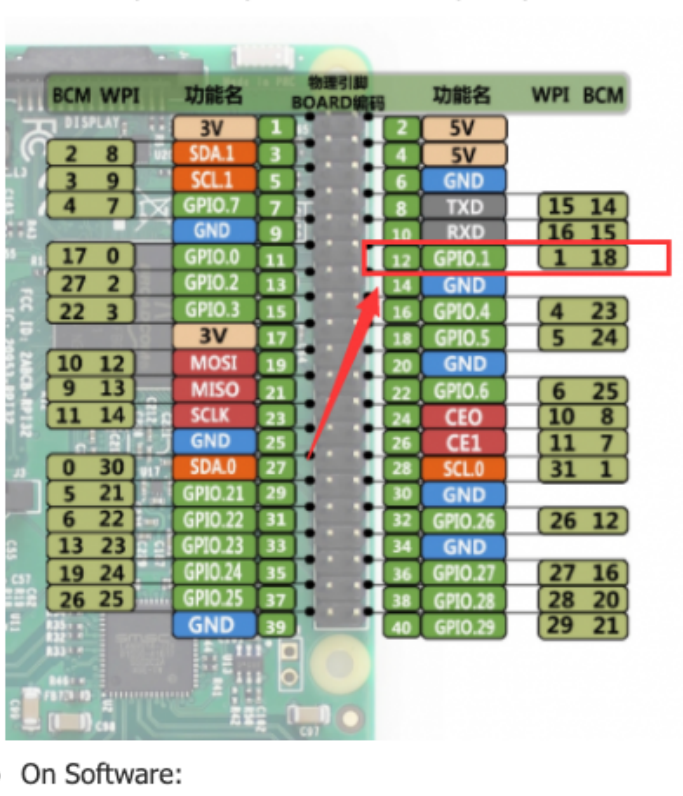
1. Please download the latest version of the image from Raspberry Pi official website.
2. Download the compressed file to the PC and extract it to get the .img file.
3. Connect the TF card to the PC, use the SDFormatter software to format the TF card.
4. Open the Win32DiskImager software, select the system image prepared in step 1, and click write to burn the system image.
5. After the programming is completed, open the config.txt file in the root directory of the TF card, add the following code at the end of config.txt, save and safely eject the TF card.

```
hdmi_group=2
hdmi_mode=87
hdmi_cvt 1024 600 60 6 0 0 0
hdmi_drive=1
```

6. Insert the TF card into the Raspberry Pi, power on the Raspberry Pi, and wait for a few seconds to display normally.

PWM Software Light Adjustment

- **Raspberry Pi bullseye system with development accessories as an example.**
- Hardware connection: Connect the PWM pin of the development accessory to the P1 pin of the Raspberry Pi (as shown in the figure below) and the GND pin to any GND of the Raspberry Pi.



- On Software:

Upgrade the Wiring Pi:

```
wget https://project-downloads.drogon.net/wiringpi-latest.deb
sudo dpkg -i wiringpi-latest.deb
gpio -v
# Run gpio -v and version 2.52 will appear. If it does not appear, the installation is wrong.
```

After the wiringPi is updated successfully, the backlight brightness can be controlled by the following commands.

```
gpio -g mode 18 pwm #Occupied pins are PWM pins
gpio pwmc 100
gpio -g pwm 18 0 #brightest
gpio -g pwm 18 1023 #darkest
gpio -g mode 18 out #Release as output
```

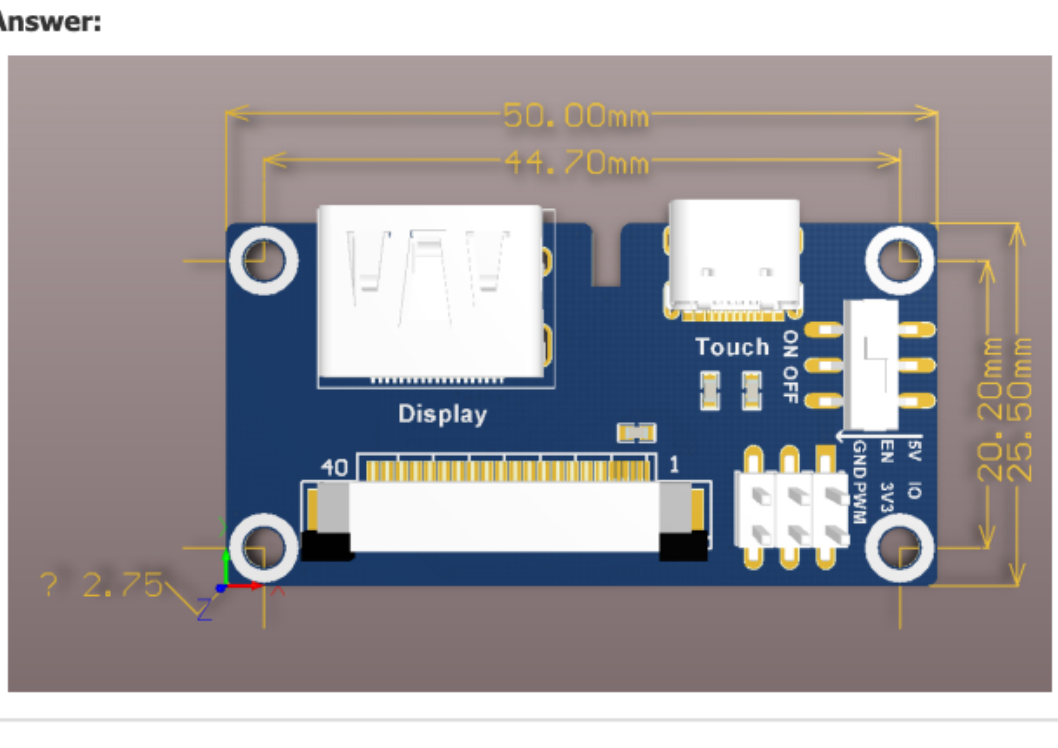
FAQ

Question: What is the power consumption of the 70H-1024600 series?

Answer:
Under the power supply of 5V, the working current is about 450mA when the backlight is on, and when the backlight is off it is about 100mA.

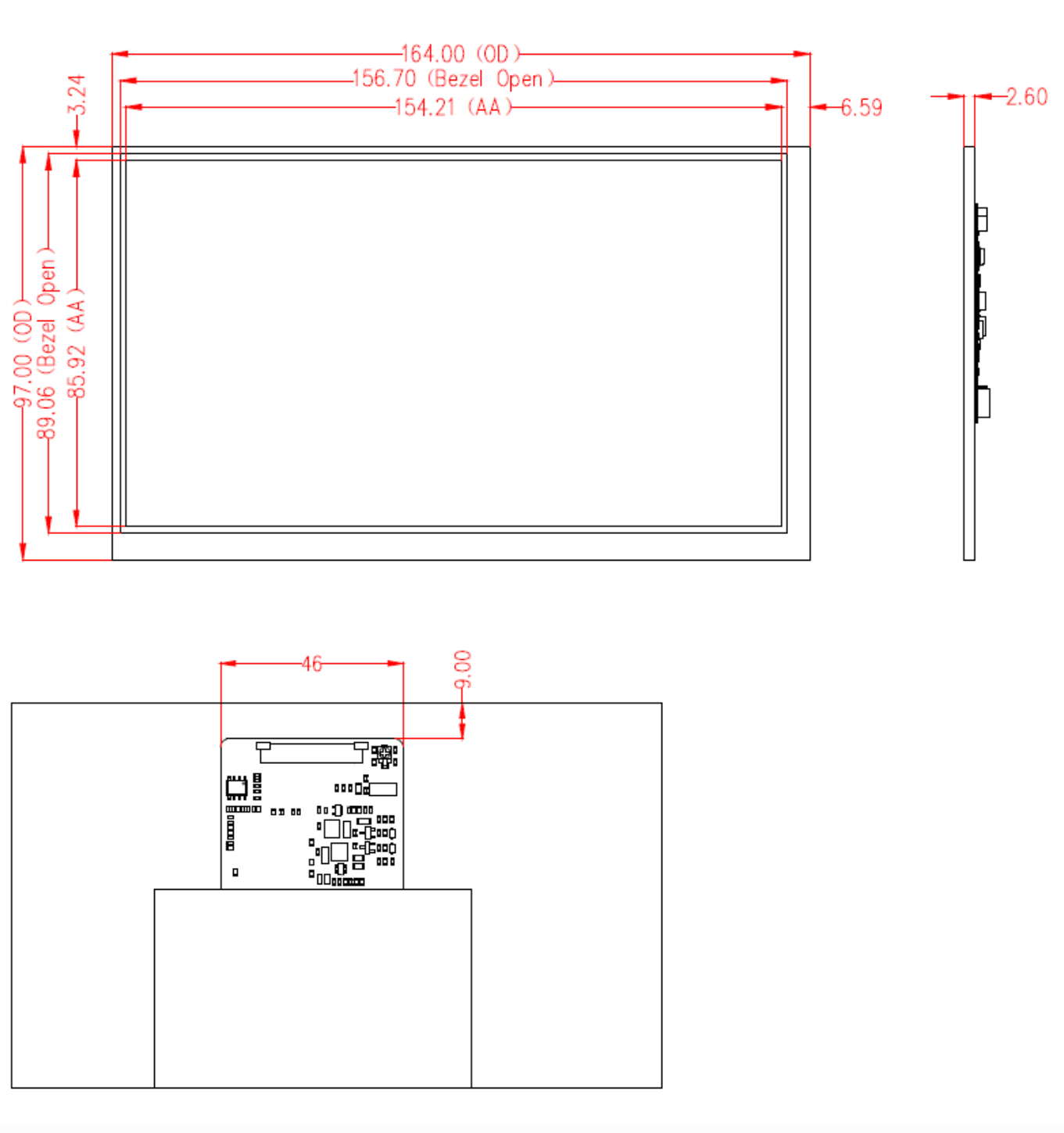
Question: Dimensional drawing of 70H-1024600 adapter board

Answer:

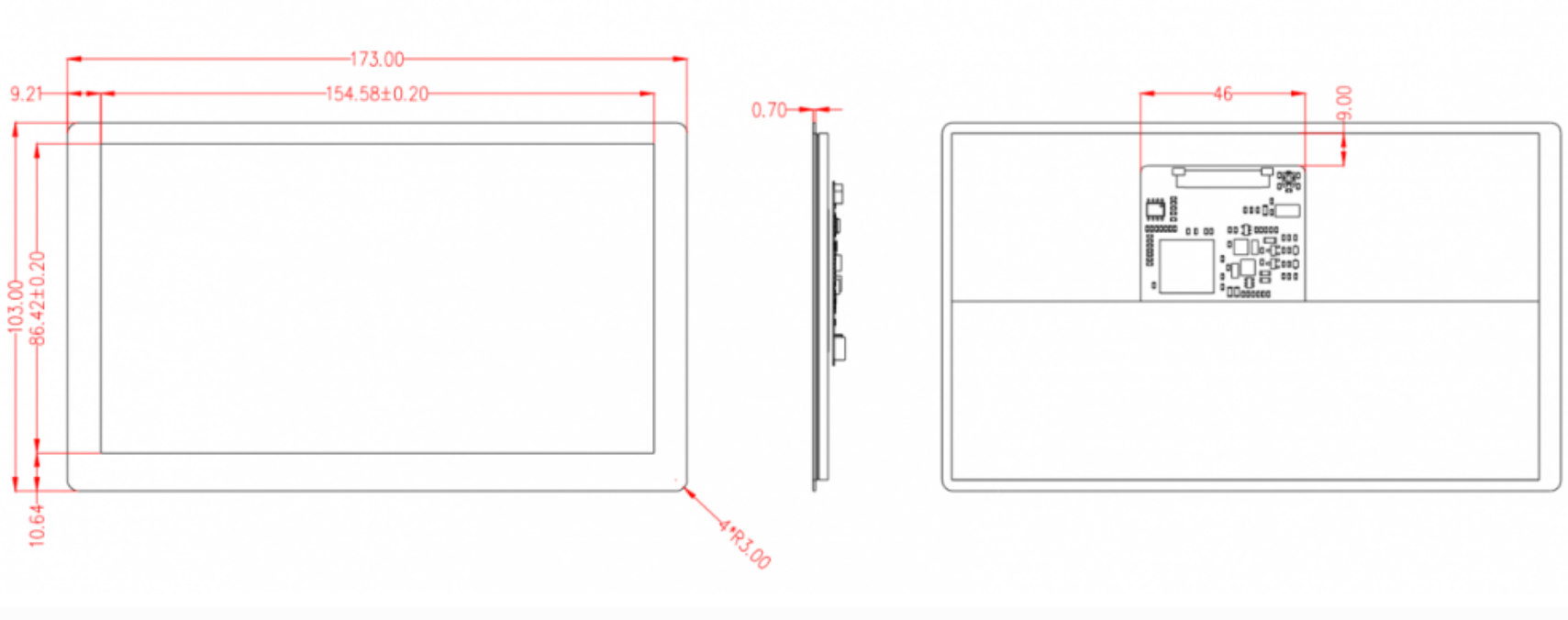


Resource

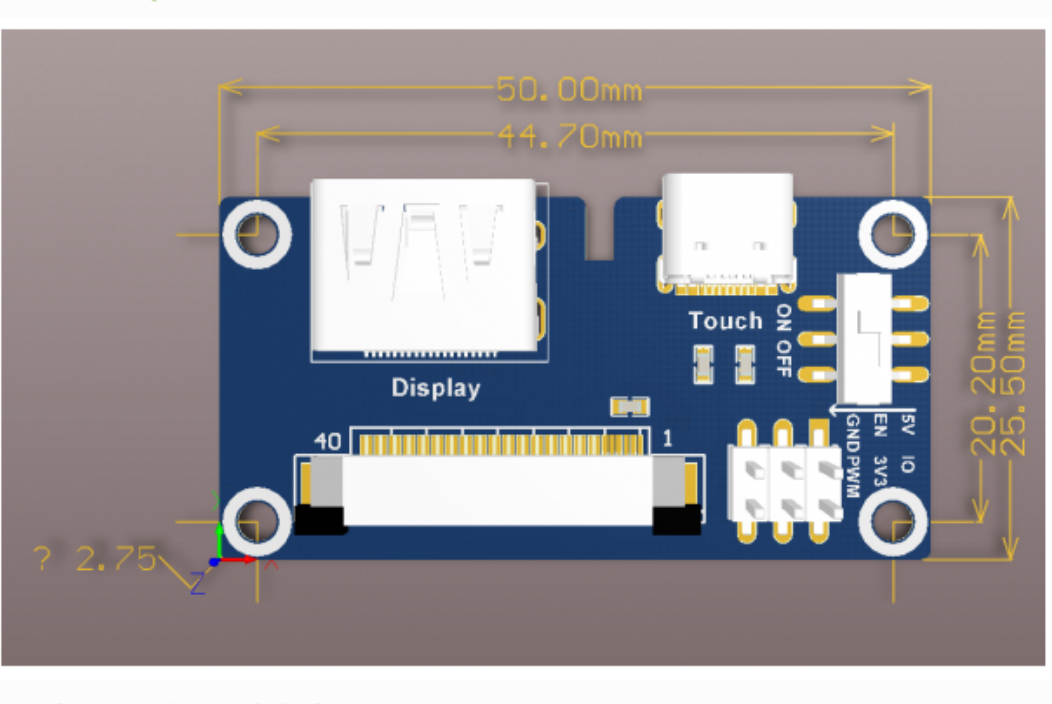
LCD Display Dimension



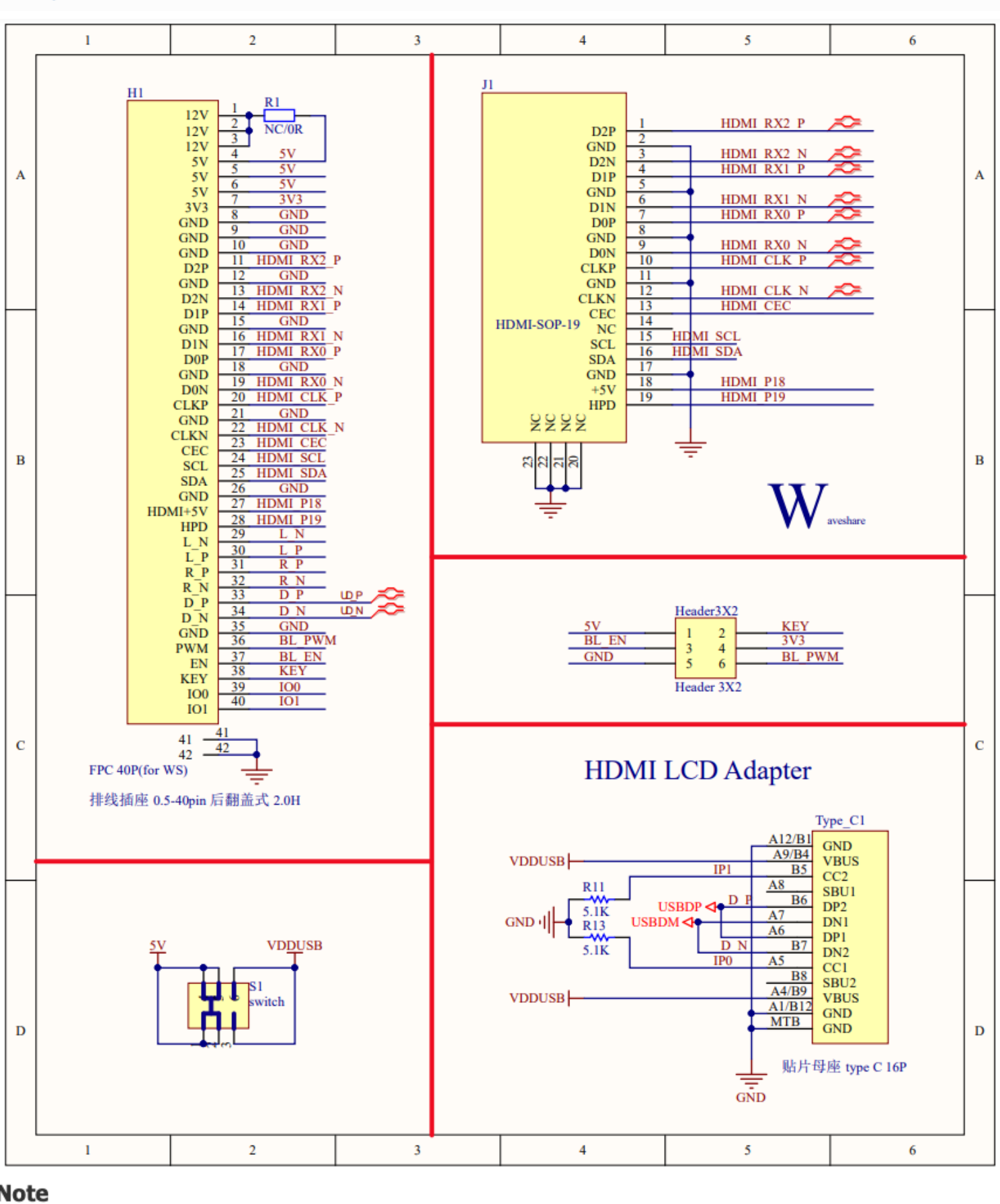
Touch Display Dimension



HDMI Apater Board Dimensions



Adapter Board Schematic



Note

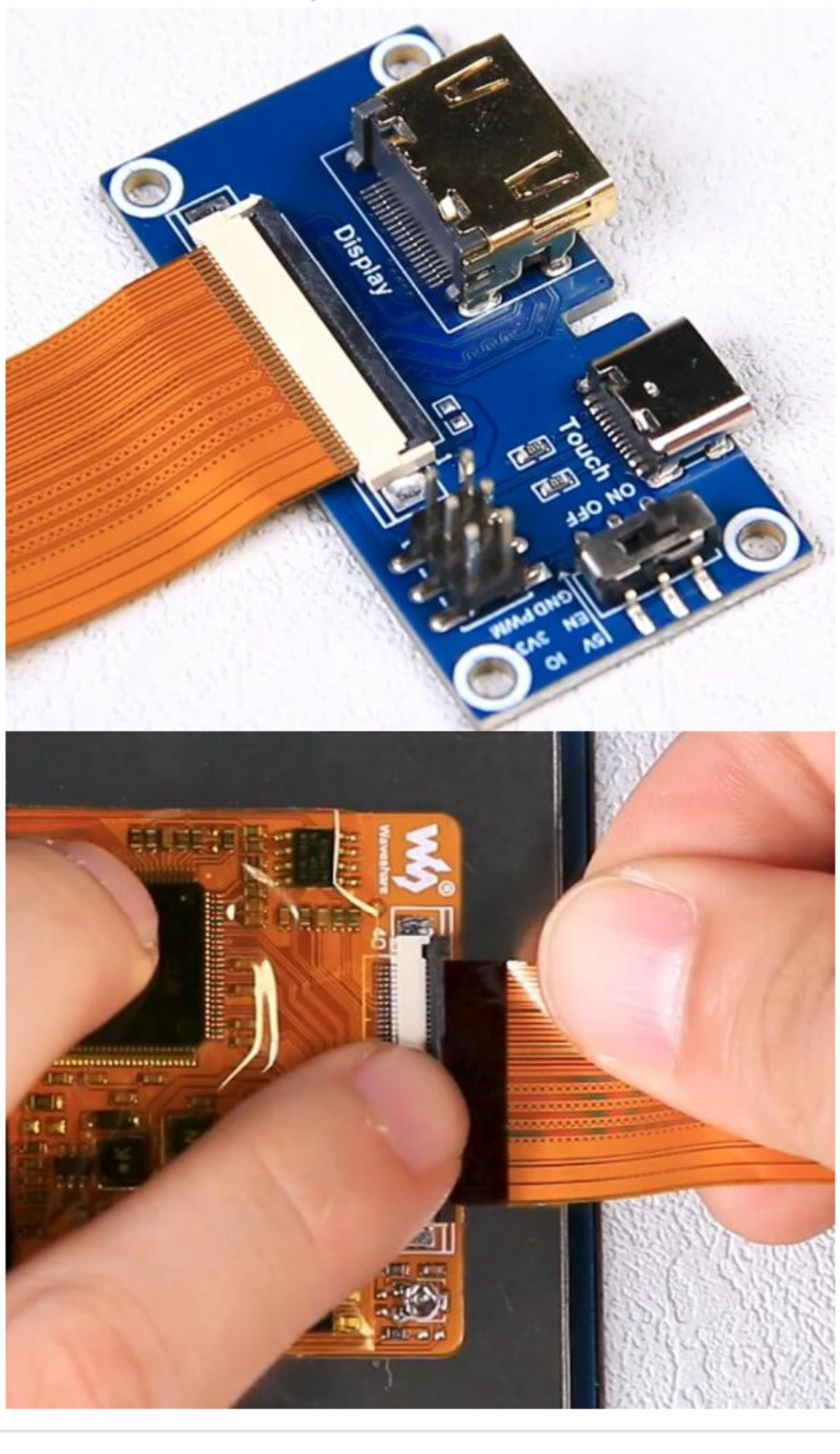
- People who handle this module should connect the display to the ground by appropriate means, and the protective film must be removed from the LCD surface very slowly to prevent static electricity from occurring.
- Do not plug or unplug the display while the power is on, otherwise the display board or main control board may be damaged.
- Please refer to the electrical parameter description to make the display screen work within the limited value range, otherwise the display screen may be damaged.
- Do not disassemble the display, as this may damage the sensitive parts inside and may cause scratches or dust on the display. Removing the display will void the warranty.
- Please do not store the display in a high temperature and high humidity environment for a long time.
- Do not touch or rub the surface of the display with anything harder than HB pencil lead.
- If the LCD is damaged, do not swallow or inhale the liquid crystal, and do not touch the liquid crystal with your skin; if the liquid crystal touches your mouth or eyes, rinse it off with water immediately; if the liquid crystal touches your skin or cloth, rinse it off with alcohol immediately, then rinse thoroughly with water; handle glass shards with care to avoid injury.

FAQ

Question: Why does the display show overcurrent protection when connecting the monitor to the computer?

Answer:

Please check whether the wiring is connected in reverse, as shown below



Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.
Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

Submit Now